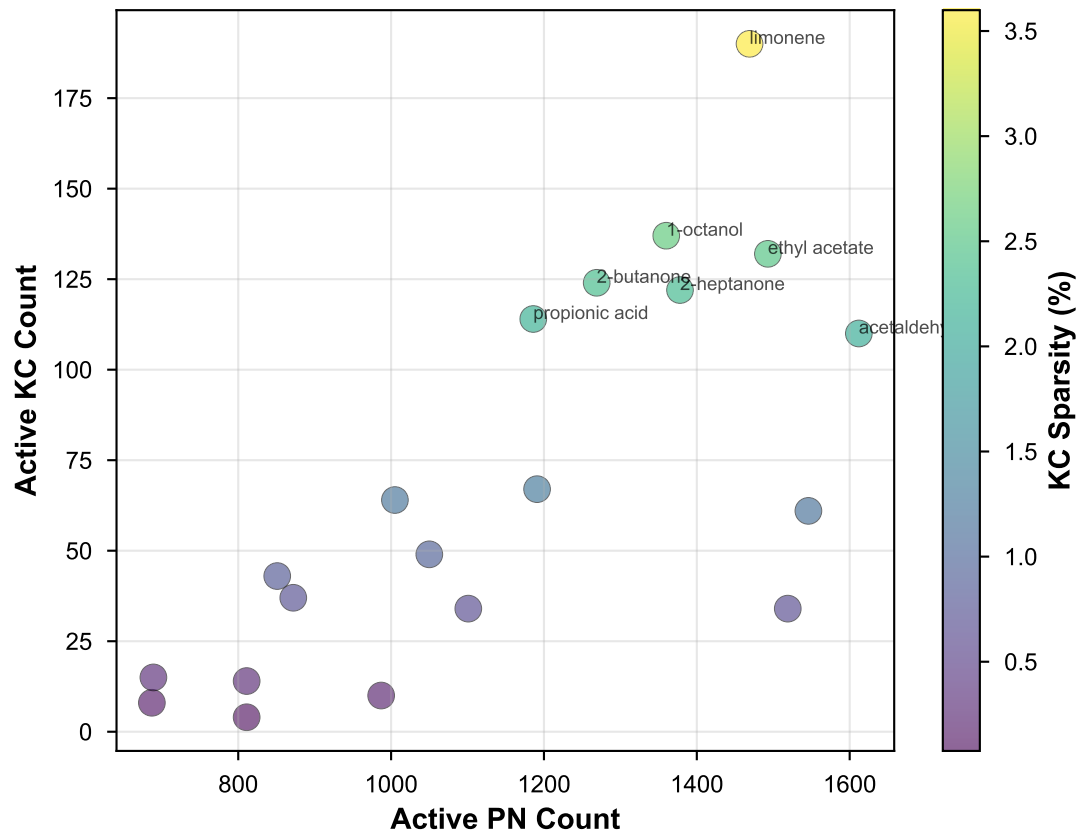


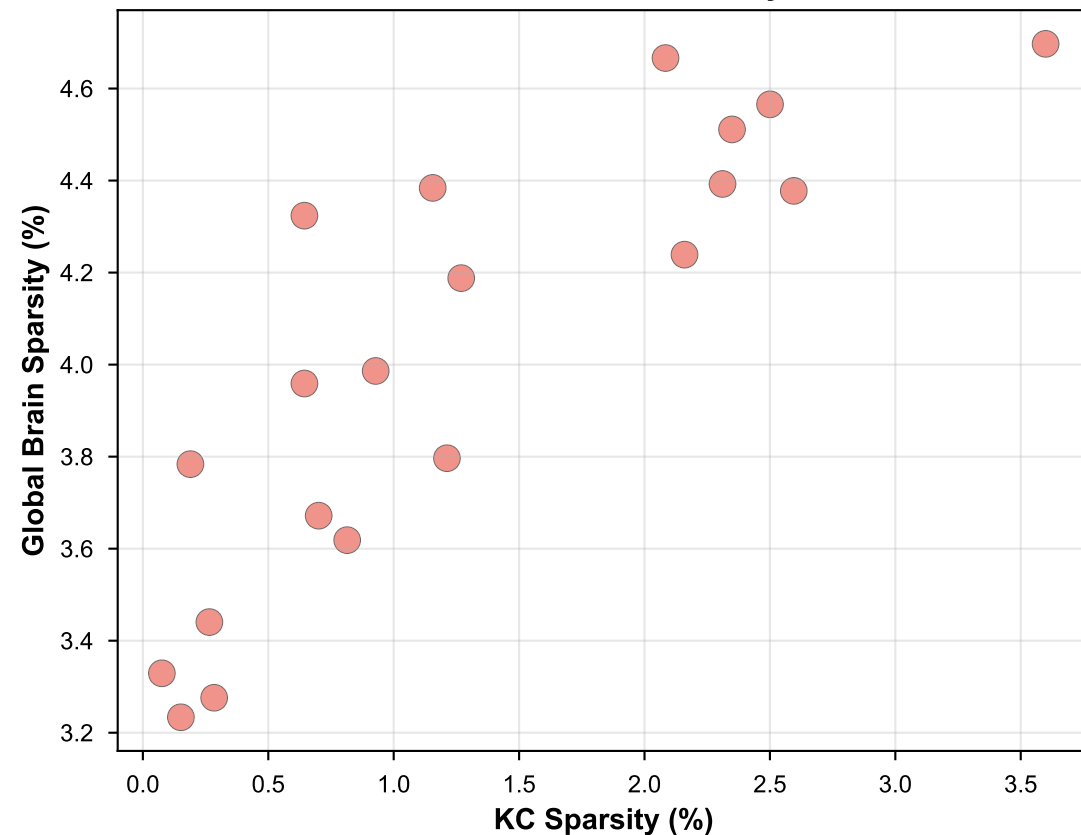
Odor-Specific Neural Response Patterns

Complete Fly Brain Wave-Based Simulation

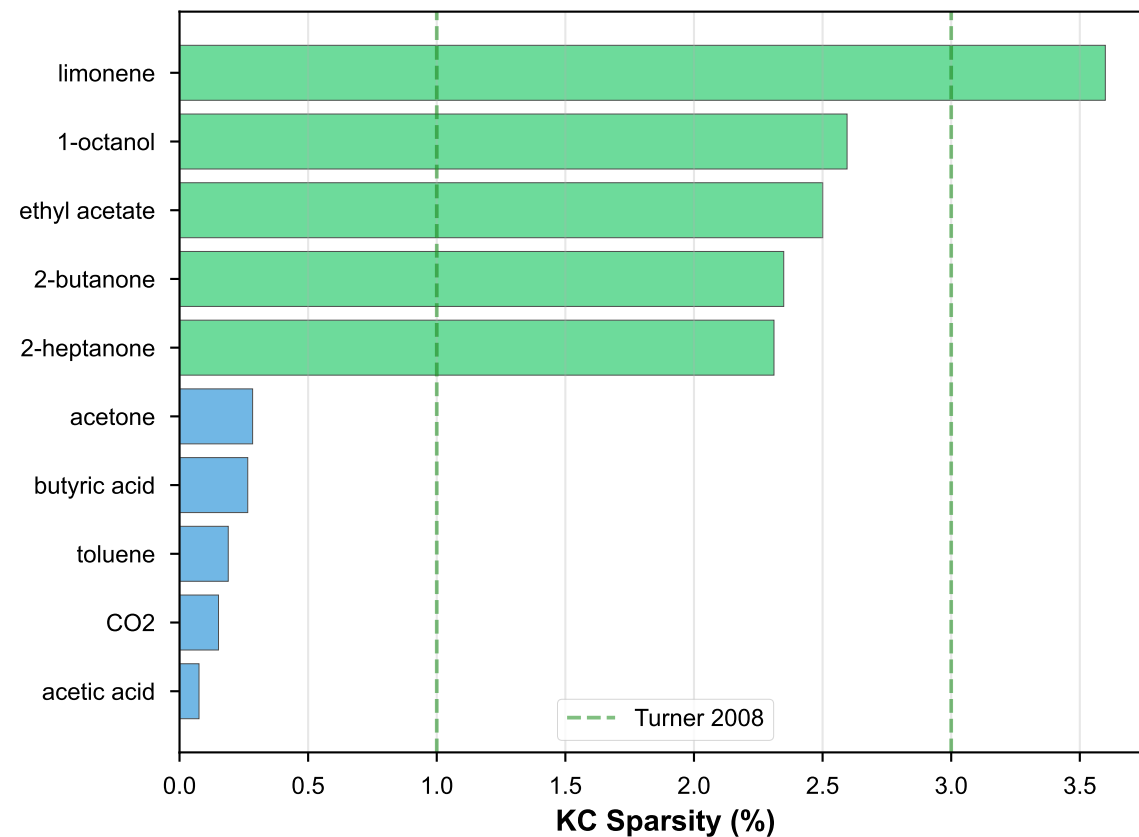
PN → KC Transformation



Local vs Global Activity



Least & Most Active Odors



SUMMARY STATISTICS (n=20 odors)

Kenyon Cell (KC) Activity:

- Mean: 68.5 neurons (1.30%)
- Median: 55 neurons (1.04%)
- Range: 4-190 neurons
- Std Dev: 52.3 neurons

Validation:

- 9/20 odors match Turner 2008 (1-3%)
- 10/20 odors in biological range (1-10%)

Global Brain:

- Mean activity: $4.02\% \pm 0.47\%$
- ~5600 neurons active per odor

Performance:

- Simulation: ~26s per 100ms (10× real-time)
- Memory: 64 MB for 139,255 neurons
- Hardware: Apple M4 Pro with MLX GPU